FIT2179 Data Visualisation
Course Notes

DATA-INK RATIO AND CHARTJUNK

Monash University
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DATA-INK RATIO

The concept of Data-int ratio was introduced by Edward Tufte in his book "The Visual Display of Quantitative Information".

TERMINOLOGY

The *data–ink ratio* refers to the proportion of a visualisation's ink that is used to represent actual data. In other words, it is defined as the ratio between two areas: the area used to encode the data and the total area of the visualisation.

The formula to calculate the data-ink ratio is presented as follows:

$$\text{data-ink ratio} = \frac{\text{ink for elements that encode data}}{\text{total ink for all elements}}$$

One way to think about the data–ink ratio is in terms of **pixels**. Consider the number of pixels used to encode data compared with the total number of pixels used to display the visualisation.

According to Tufte, the data–ink ratio should ideally be **as close to 1 as possible**. This means that most of the visual elements in a chart should be devoted to displaying data rather than decorative or unnecessary elements.

CHARTJUNK

A related concept is chartjunk, which refers to visual clutter that does not contribute to understanding the data.

TERMINOLOGY

C*hartjunk* refers to all ornamental elements added to the visualisation.

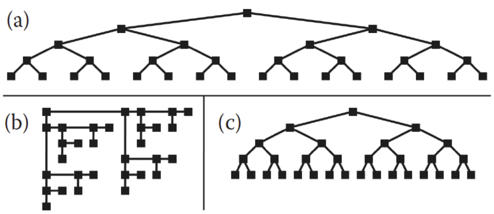
In this unit, we aim to reduce or remove chartjunk from visualisations. This means stripping away visual elements that are not dedicated to representing data. By doing this, we can increase the data–ink ratio and focus attention on the information being presented.

The concepts of data–ink ratio and chartjunk help us think about how to design clearer visualisations and increase the information density of charts.

TERMINOLOGY

The term *information density* is commonly used in the data analysis and visualisation literature. It refers to the amount of information encoded relative to the amount of unused space in a visualisation.

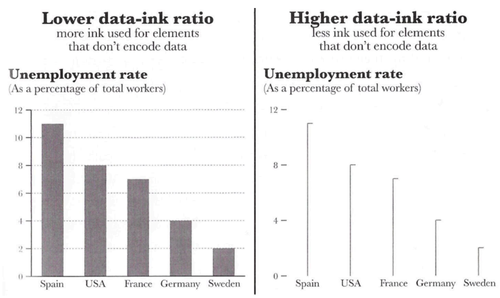
For example, visualisation (a) has low information density, as a large amount of space is used to display relatively little data. Visualisations (b) and (c), in contrast, have higher information density, displaying more data within the reduce visual space.



However, Tufte’s strong criticism of visual clutter is not always universally accepted. While good visualisation design aims to remove unnecessary elements, there are limits to how much can be removed. In some cases, maximising the data–ink ratio does not necessarily produce the most effective visualisation.

Research has shown that highly minimalist visualisations may not always support understanding. In some situations, additional visual elements can make charts easier to interpret.

For example, a 2007 study from Ben-Gurion University presented 87 students with traditional bar charts and charts designed to maximise the data–ink ratio. Many participants rejected the minimalist version, possibly because the traditional bar chart format was more familiar.



Furthermore, some studies suggest that ornamental elements can increase memorability or create a more positive user experience when interacting with a visualisation.

In summary, data visualisation designers should aim to remove unnecessary or redundant visual elements that do not support interpretation. However, visualisations should also remain clear, interpretable, and engaging, even if that means including some non-data elements.

FURTHER READING

Edward Tufte (1983). The Visual Display of Quantitative Information.

Minimalism in information visualization: attitudes towards maximi ing the data-ink ratio (2007). Onhad In-bar, Noam Tractinsky, Joachim Meyer

Scott Bateman, Regan L. Mandryk, Carl Gutwin, Aaron Genest, David McDine, Christopher Brooks (2010). Useful Junk? The Effects of Visual Embellishment on Comprehension and Memorability of Charts.